

# Minimal winding in the self-similar collapse of three point vortices and of two concentric vortex polygons

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## Abstract

In a self-similar collapse of point vortices every vortex moves on a logarithmic spiral, and the dimensionless number  $P = |\omega_0|t_c$ , the initial angular velocity times the collapse time, measures how tightly the spiral winds. We minimize  $P$  over two classical collapsing families. Every self-similar collapse of three point vortices can be normalized to circulations  $(1, \mu, -\mu/(1+\mu))$  with  $0 < \mu \leq 1$  and zero angular impulse. For each  $\mu$  the collapsing configurations form two arcs, one for each orientation of the vortex triangle, and  $P$  has exactly one critical point, a minimum, on each. The squares of the two minima are roots of an explicit cubic whose coefficients are polynomials in  $\mu$ , and for  $\mu < 1$  the two minima differ. The smaller one increases strictly from  $\sqrt{3}/2$ , approached as  $\mu \rightarrow 0$ , to  $\sqrt{2}$  at  $\mu = 1$ . Hence  $P > \sqrt{3}/2$  for every self-similar collapse of three point vortices, and the constant is sharp; equivalently, every vortex travels more than twice its initial distance from the collision point. For  $\mu = 1/2$  the two minima are  $1.0647059762\dots$  and  $2.2038550160\dots$ , the positive roots of  $8748\xi^6 - 49005\xi^4 + 27794\xi^2 + 18723$ , and they are not expressible by real radicals. For two concentric regular  $n$ -gons with circulations  $x_n$  and  $-1$ ,  $P = (K_n - \sqrt{2n-1} \cos n\theta)/(2n \sin n\theta)$  in terms of the relative rotation  $\theta$ , with an explicit constant  $K_n$ , and the minimum over  $\theta$  is  $\sqrt{K_n^2 - 2n + 1}/(2n)$ ; for pentagons it is  $\sqrt{31682}/80$ . All formulas are also checked against the Biot–Savart velocities in high-precision arithmetic.

**Keywords:** point vortices; vortex collapse; self-similar motion; logarithmic spiral.

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## 1 Introduction

Three point vortices collapse self-similarly only if  $1/\Gamma_1 + 1/\Gamma_2 + 1/\Gamma_3 = 0$  and their angular impulse about the center of vorticity vanishes (Lemma 2 below). Under these conditions every configuration moves self-similarly: the triangle keeps its shape and rotates while it shrinks to a point in finite time, expands, or rotates rigidly [9, Sect. 4.5.2], [17]. Such collapse was found by Gröbli [11] and found again by Aref [1] and by Novikov and Sedov [21], who also constructed collapsing configurations of four and five vortices; see [4] for the history and [20] for the general theory. Synge [24, Sect. 4, Theorem 8] identified  $\Gamma_1\Gamma_2 + \Gamma_2\Gamma_3 + \Gamma_3\Gamma_1 = 0$  as an exceptional case: the configurations whose shape lies on a certain hyperbola in his trilinear coordinates then come in one-parameter families of similar triangles, so that the shape does not determine the size. Conte and de Seze [6] solved the motion of three vortices of arbitrary circulations exactly and showed that when  $\sum_{j < k} \Gamma_j \Gamma_k = 0$  and the angular impulse vanishes, each vortex runs a logarithmic spiral about the center of vorticity while the triangle keeps its shape, and the configuration either collapses in finite time or expands [6, Sect. 4]. Kimura [15] studied similarity solutions of point-vortex systems, Aref [3] derived formulas for the rate of collapse or expansion and the angular frequency of rotation, and Gotoda [10] gave

explicit formulas for the self-similar motions of three vortices. Borisov and Lebedev [5] obtained conditions for the collapse and the scattering of three vortices within the Lie–Poisson formulation of the problem, and Krishnamurthy, Aref and Stremler [17] recast the motion of three vortices in terms of the circumcircle and the interior angles of the vortex triangle and derived equations of motion for these quantities. Collapse also occurs in configurations of higher symmetry. Aref [2] reduced the motion of two concentric regular  $n$ -gons of vortices to an integrable Hamiltonian system with two degrees of freedom, and Koiller et al. [16] found collapsing configurations of two such rings, whose vortices move on logarithmic spirals. Demina and Kudryashov [7, Sect. 3] gave explicitly a family of such configurations that collapse or scatter, with an additional vortex, possibly of zero circulation, at the center, together with the complex constant that determines their rates of collapse and rotation.

In a self-similar collapse at time  $t_c$  the configuration has rotated by time  $t$  through the angle  $-\omega_0 t_c \ln(1 - t/t_c)$ , where  $\omega_0$  is the initial angular velocity. The angular velocity at time  $t$  is  $\omega_0 t_c / (t_c - t)$ , so the product of the angular velocity and the remaining time does not depend on which instant is taken as initial. Each vortex moves on a logarithmic spiral about the collision point, whose exponent is fixed by this product [3, Eq. (29)], and the dimensionless number  $P = |\omega_0| t_c$ , which is invariant under rescaling of lengths, times and circulations, is the angle through which the configuration turns while the square of its size decreases by the factor  $e$ . Equivalently, the path of each vortex makes the constant angle  $\arctan 2P$  with the direction to the collision point, and a vortex that starts at distance  $r_0$  from that point travels the distance  $r_0 \sqrt{1 + 4P^2}$  before the collapse. Within a collapsing family it is natural to ask which configuration winds least.

For three vortices we answer this question completely (Section 3). After normalization the circulations are  $(1, \mu, -\mu/(1 + \mu))$  with  $0 < \mu \leq 1$ . The collapsing configurations form two arcs, one for each orientation of the triangle, and  $P$  has a unique minimum on each (Theorem 1). For  $\mu < 1$  the two minima differ, and the smaller one increases strictly with  $\mu$ , from  $\sqrt{3}/2$  as  $\mu \rightarrow 0$  to  $\sqrt{2}$  at  $\mu = 1$ , the value for two equal circulations. Thus  $P > \sqrt{3}/2$  for every self-similar collapse of three vortices, and the bound is sharp (Corollary 1). For  $\mu = 1/2$  the two minima are conjugate algebraic numbers of degree six, given explicitly in Proposition 1. In Section 4 we treat two concentric regular  $n$ -gons, for which the minimization reduces to an elementary inequality once  $P$  is written in a suitable form. Section 5 describes an independent numerical check.

## 2 Self-similar collapse

The positions  $z_j \in \mathbb{C}$  of point vortices with circulations  $\Gamma_j$  evolve according to

$$\dot{z}_j = \frac{1}{2\pi i} \sum_{k \neq j} \frac{\Gamma_k}{z_j - z_k}. \quad (1)$$

Suppose  $\sum_j \Gamma_j \neq 0$ , let  $z_c = \sum_j \Gamma_j z_j / \sum_j \Gamma_j$  be the center of vorticity, which is conserved, and suppose that at  $t = 0$  there is  $\kappa \in \mathbb{C}$  with  $\dot{z}_j = \kappa(z_j - z_c)$  for all  $j$ . The right-hand side of (1) is homogeneous of degree  $-1$  in the relative positions and equivariant under rotations, so the solution is  $z_j(t) = z_c + \lambda(t)e^{i\phi(t)}(z_j(0) - z_c)$  with  $\lambda\dot{\lambda} = \operatorname{Re} \kappa$  and  $\lambda^2\dot{\phi} = \operatorname{Im} \kappa$ . Hence  $\lambda^2 = 1 + 2t \operatorname{Re} \kappa$ . If  $\operatorname{Re} \kappa < 0$  the vortices collide at  $z_c$  at the time  $t_c = -1/(2 \operatorname{Re} \kappa)$ , and  $\omega_0 = \operatorname{Im} \kappa$ , so that

$$P = |\omega_0| t_c = \frac{|\operatorname{Im} \kappa|}{-2 \operatorname{Re} \kappa}. \quad (2)$$

Integrating  $\dot{\phi} = \omega_0 / \lambda^2$  gives  $\phi = -P \ln \lambda^2$  when  $\omega_0 > 0$ , which is the description of the spiral given in the introduction. This is Kimura’s similarity solution [15, Sect. 3.1]: up to the factor  $2\pi$  in his

normalization of the circulations,  $\kappa$  is his constant  $C = A + iB$ , (2) reads  $P = |B|/(-2A)$ , and the spirals and the collision time  $t_c = -1/(2A)$  are his Eqs. (3.7) and (3.9). Earlier, Conte and de Seze [6, Sect. 4, pp. 24–25] wrote this motion as  $z_j = z_{j,0}(1 - t/t_c)^{1/2 - i\omega t_c}$  and gave  $-2\omega + i/t_c$  in closed form in their shape variable; there  $\omega$  is the initial angular velocity, so  $|\omega t_c| = P$ . Demina and Kudryashov [7, Eqs. (7)–(12)] derive the same solution, like Kimura for any number of vortices but with the normalization of (1), so that their complex constant  $c_1$  equals  $\kappa$ ; they give the collapse time  $1/(2|\operatorname{Re} c_1|)$  [7, Eq. (12)] and the rotation angle  $(\operatorname{Im} c_1/(2\operatorname{Re} c_1)) \ln(1 + 2t\operatorname{Re} c_1)$  [7, Eq. (10)], whose coefficient has absolute value  $P$ . Galloway and Šverák [9, Prop. 5.2, Eqs. (5.5), (5.11)] write the collapsing solution as  $z_j(t) = (1 - t/T)^{1/2 + is} a_j$  with  $z_c = 0$ ; comparing the two descriptions gives  $T = t_c$  and  $s = -\omega_0 t_c$ , so  $|s| = P$ . The angular impulse about the center of vorticity satisfies

$$\sum_j \Gamma_j |z_j - z_c|^2 = \frac{1}{\sum_j \Gamma_j} \sum_{j < k} \Gamma_j \Gamma_k |z_j - z_k|^2. \quad (3)$$

It is conserved and, in a self-similar motion, proportional to  $\lambda^2$ , so it must vanish for a collapse. Two of the minimizations below (Remark 2 and Proposition 2) reduce to the following elementary inequality.

**Lemma 1.** *Let  $a > |b|$ . For  $0 < \alpha < \pi$ ,  $(a - b \cos \alpha)/\sin \alpha \geq \sqrt{a^2 - b^2}$ , with equality if and only if  $\cos \alpha = b/a$ .*

*Proof.* The numerator is positive, and  $(a - b \cos \alpha)^2 - (a^2 - b^2) \sin^2 \alpha = (a \cos \alpha - b)^2$ . □

### 3 Three vortices

#### 3.1 Normalization and the collapsing configurations

The Hamiltonian  $H = -(4\pi)^{-1} \sum_{j < k} \Gamma_j \Gamma_k \ln |z_j - z_k|^2$  of (1) is conserved. We call a collapse self-similar if all mutual distances are  $\lambda(t)$  times their initial values, with  $\lambda(t) \rightarrow 0$  in finite time.

**Lemma 2.** *In a self-similar collapse of three point vortices all circulations are nonzero,  $1/\Gamma_1 + 1/\Gamma_2 + 1/\Gamma_3 = 0$ ,  $\sum_j \Gamma_j \neq 0$ , and the angular impulse about the center of vorticity vanishes.*

*Proof.* The harmonic condition and the vanishing of the angular impulse are the classical necessary conditions [3, Sect. II B] (see also [15, Eqs. (3.20)–(3.21), (3.33)], [5, Sect. 3] and, for both conditions and any number of vortices, [7, Eqs. (26)–(28)]); we include the short argument. Along the motion

$$H = H(0) - \frac{1}{4\pi} \left( \sum_{j < k} \Gamma_j \Gamma_k \right) \ln \lambda^2,$$

and  $\ln \lambda^2 \rightarrow -\infty$ , so  $\sum_{j < k} \Gamma_j \Gamma_k = 0$ . If one circulation vanished, this would force a second one to vanish; the remaining vortex would then be at rest and the other two would move on circles about it, so no collapse would occur. Hence all  $\Gamma_j \neq 0$ , and  $\sum_{j < k} \Gamma_j \Gamma_k = \Gamma_1 \Gamma_2 \Gamma_3 \sum_j 1/\Gamma_j$  gives the harmonic condition. Moreover  $(\sum_j \Gamma_j)^2 = \sum_j \Gamma_j^2 > 0$ . Finally, the angular impulse (3) is conserved and proportional to  $\lambda^2$ , so it vanishes. □

By Lemma 2 two of the three circulations of a self-similarly collapsing configuration have the same sign. Multiplying all circulations by a positive constant rescales time and leaves  $P$  unchanged. Multiplying them by  $-1$  reverses time, and so does complex conjugation of the positions, which

maps solutions of (1) to solutions; the composition maps collapsing solutions to collapsing solutions with the same  $P$ . After relabeling we may therefore assume, as Gotoda does [10, Sect. 3],

$$\Gamma = \left(1, \mu, -\frac{\mu}{1+\mu}\right), \quad 0 < \mu \leq 1. \quad (4)$$

Then  $\sum_j \Gamma_j = R/(1+\mu) > 0$ , where  $R = 1 + \mu + \mu^2$ . For  $w = (z_3 - z_1)/(z_2 - z_1)$  the sum  $\sum_{j < k} \Gamma_j \Gamma_k |z_j - z_k|^2$  in (3) equals  $\mu(1+\mu)^{-1}|z_2 - z_1|^2(1 + 2\mu \operatorname{Re} w - (1+\mu)|w|^2)$ , which vanishes exactly on the circle  $|w - \mu/(1+\mu)| = \sqrt{R}/(1+\mu)$ . This circle appears in Kimura [15, Eqs. (3.34)–(3.35)], who notes that it splits into two arcs of collapse and two of expansion, and in Aref [3, Eq. (20)]; we use the parametrization of it given by Gotoda [10, Sect. 3]:

$$z_1 = \frac{\mu(1 + \sqrt{R}e^{-i\theta})}{(1+\mu)^2}, \quad z_2 = \frac{\mu - \sqrt{R}e^{-i\theta}}{(1+\mu)^2}, \quad z_3 = 1, \quad \theta \in [0, 2\pi). \quad (5)$$

Here  $\sum_j \Gamma_j z_j = 0$ , so  $z_c = 0$ , and

$$z_1 - z_2 = \frac{\sqrt{R}e^{-i\theta}}{1+\mu}, \quad z_3 - z_1 = \frac{\sqrt{R}(\sqrt{R} - \mu e^{-i\theta})}{(1+\mu)^2}, \quad z_3 - z_2 = \frac{\sqrt{R}(\sqrt{R} + e^{-i\theta})}{(1+\mu)^2}. \quad (6)$$

The shape ratio is  $w = \mu/(1+\mu) - (\sqrt{R}/(1+\mu))e^{i\theta}$ , which traverses the zero-impulse circle exactly once, so every zero-impulse configuration is obtained, up to translation, rotation and dilation, for exactly one  $\theta$ . Since  $\operatorname{Im} w = -(\sqrt{R}/(1+\mu))\sin \theta$ , the triangle  $z_1 z_2 z_3$  is negatively oriented for  $\sin \theta > 0$  and positively oriented for  $\sin \theta < 0$ . Put  $C = \sqrt{R} \cos \theta$  and

$$N(C) = 2(1 + \mu^2)R + (1 - \mu)(2 + \mu + 2\mu^2)C - 2\mu C^2, \quad M(C) = 1 - \mu + 2C.$$

**Lemma 3.** *For every  $\theta$  the three quotients  $\dot{z}_j/(z_j - z_c)$  are equal to*

$$\kappa = \frac{i(1+\mu)^3}{2\pi\sqrt{R}} \cdot \frac{\sqrt{R} + (1-\mu)e^{i\theta}}{(\sqrt{R} - \mu e^{i\theta})(\sqrt{R} + e^{i\theta})},$$

and, with  $D = (R + \mu^2 - 2\mu C)(R + 1 + 2C) > 0$ ,

$$\operatorname{Re} \kappa = -\frac{(1+\mu)^3 \mu M(C) \sin \theta}{2\pi\sqrt{R} D}, \quad \operatorname{Im} \kappa = \frac{(1+\mu)^3 N(C)}{2\pi R D}.$$

Moreover  $N(C) > 0$  for every  $\theta$ .

*Proof.* Since  $z_c = 0$  and  $z_3 = 1$ ,  $\kappa = \dot{z}_3$ , and by (1),  $\bar{\dot{z}}_3 = (2\pi i)^{-1}((z_3 - z_1)^{-1} + \mu(z_3 - z_2)^{-1})$ . Inserting (6) gives the stated  $\kappa$ . The same computation for  $j = 1$  and  $j = 2$  gives the same value, as it must, since these configurations move self-similarly [15], [18, Sect. 3.1]. The real and imaginary parts follow on multiplying numerator and denominator by the complex conjugate of the denominator, since  $|\sqrt{R} - \mu e^{i\theta}|^2 |\sqrt{R} + e^{i\theta}|^2 = D$ . The two factors of  $D$  are at least  $(\sqrt{R} - \mu)^2$  and  $(\sqrt{R} - 1)^2$ , which are positive because  $R > \mu^2$  and  $R > 1$ . Finally,  $N$  is concave in  $C$ , and  $N(\sqrt{R})N(-\sqrt{R}) = 3\mu^2(1+\mu)^2 R$  and  $N(\sqrt{R}) + N(-\sqrt{R}) = 4R(1 - \mu + \mu^2)$  are positive, so  $N > 0$  on  $|C| \leq \sqrt{R}$ .  $\square$

By Lemma 3 every zero-impulse configuration rotates in the positive sense, and it collapses exactly when  $M(C) \sin \theta > 0$ . Let  $\theta_0 \in (0, \pi)$  be defined by  $\cos \theta_0 = (\mu - 1)/(2\sqrt{R})$ . The configurations

with  $\theta = 0$  and  $\theta = \pi$  are collinear, and those with  $\theta = \pm\theta_0$  are equilateral triangles; these four are relative equilibria. The collapsing configurations form the two arcs

$$\mathcal{A}_+ = (0, \theta_0), \quad \mathcal{A}_- = (\pi, 2\pi - \theta_0),$$

the triangle being negatively oriented on  $\mathcal{A}_+$  and positively oriented on  $\mathcal{A}_-$ , and on them (2) gives

$$P(\theta) = \frac{N(C)}{2\mu\sqrt{R}M(C)\sin\theta}, \quad C = \sqrt{R}\cos\theta. \quad (7)$$

Since  $z_j(-\theta) = \overline{z_j(\theta)}$ , the reflection  $\theta \mapsto -\theta$  maps  $\mathcal{A}_+$  and  $\mathcal{A}_-$  onto the expanding arcs  $(2\pi - \theta_0, 2\pi)$  and  $(\theta_0, \pi)$ .

### 3.2 The minimal winding

Let  $u = \mu + 1 + 1/\mu \geq 3$  and  $Q(\mu, y) = \mu^6 \tilde{Q}(u, y)$ , where

$$\begin{aligned} \tilde{Q}(u, y) = & 1728(u+1)^2 y^3 - 144(u+1)(8u^3 - 9u - 9)y^2 \\ & - 4(16u^6 - 288u^4 - 288u^3 - 81u^2 - 162u - 81)y + 3(4u^3 - 3u - 3)^2. \end{aligned} \quad (8)$$

Then  $Q$  is a polynomial in  $\mu$  and  $y$  with integer coefficients, irreducible over  $\mathbb{Q}$ . Its discriminant with respect to  $y$  is

$$28311552\mu^4(\mu-1)^2(\mu+1)^4(\mu+2)^2(2\mu+1)^2R^6B^3,$$

where  $B = 4\mu^6 + 12\mu^5 + 51\mu^4 + 82\mu^3 + 51\mu^2 + 12\mu + 4$ ; the sum of the roots of  $Q(\mu, \cdot)$  is  $(8u^3 - 9u - 9)/(12(u+1)) > 0$ , and their product is  $-((4u^3 - 3u - 3)/(24(u+1)))^2 < 0$ . Hence for  $0 < \mu < 1$  the cubic  $Q(\mu, \cdot)$  has three distinct real roots, one negative and two positive, which we denote  $y_1(\mu) < y_2(\mu)$ .

**Theorem 1.** *Let  $0 < \mu \leq 1$ .*

(a) *On each of  $\mathcal{A}_+$  and  $\mathcal{A}_-$  the function  $P$  has exactly one critical point, and it is the minimum of  $P$  on that arc. Denote the two minima by  $P_+(\mu)$  and  $P_-(\mu)$ .*

(b)  *$P_+(\mu)^2$  and  $P_-(\mu)^2$  are roots of  $Q(\mu, \cdot)$ . For  $\mu < 1$ ,  $P_-(\mu)^2 = y_1(\mu)$  and  $P_+(\mu)^2 = y_2(\mu)$ , so  $P_-(\mu) < P_+(\mu)$ ; for  $\mu = 1$ ,  $P_+ = P_- = \sqrt{2}$ .*

(c) *The least value of  $P$  over the collapsing configurations,  $P_-(\mu)$ , is strictly increasing on  $(0, 1]$ , and  $P_-(\mu) \rightarrow \sqrt{3}/2$  as  $\mu \rightarrow 0^+$ .*

*Proof.* (a) A direct computation from (7) gives

$$\frac{dP}{d\theta} = -\frac{G(C)}{2\mu M(C)^2 \sin^2 \theta},$$

where

$$G(C) = 4(1 - \mu)C^3 + 4(2\mu^2 - \mu + 2)C^2 + 2(1 - \mu)^3C - (2\mu^4 + 7\mu^3 + 6\mu^2 + 7\mu + 2). \quad (9)$$

As  $\theta$  runs over  $\mathcal{A}_+$ ,  $C$  decreases monotonically from  $\sqrt{R}$  to  $C_0 = (\mu - 1)/2$ ; over  $\mathcal{A}_-$  it increases from  $-\sqrt{R}$  to  $C_0$ . The critical points on the two arcs therefore correspond to the roots of  $G$  in  $(C_0, \sqrt{R})$  and in  $(-\sqrt{R}, C_0)$ . Now  $G(\pm\sqrt{R}) = \pm\sqrt{R}N(\pm\sqrt{R})M(\pm\sqrt{R})/R$ , which is positive because  $M(\sqrt{R}) > 0 > M(-\sqrt{R})$ , and  $G(C_0) = -3(1 + \mu)^4/2 < 0$ . So  $G$  has a root in each interval. For  $\mu < 1$ ,  $G$  is a cubic with positive leading coefficient, so it also has a root below  $-\sqrt{R}$ , and each

interval contains exactly one root. For  $\mu = 1$ ,  $G = 12C^2 - 24$ , with one root in each of  $(-\sqrt{3}, 0)$  and  $(0, \sqrt{3})$ . The function  $P$  is differentiable on each arc, and at the ends of each arc  $M(C) \sin \theta \rightarrow 0$  while  $N > 0$ , so  $P \rightarrow +\infty$ ; hence the single critical point is the minimum.

(b) At a critical point  $G(C) = 0$ , and since  $R \sin^2 \theta = R - C^2$ , the number  $y = P^2$  satisfies  $4\mu^2 y(R - C^2)M(C)^2 = N(C)^2$ . The resultant of these two polynomials with respect to  $C$  is  $-48\mu^4(1 + \mu)^8 Q(\mu, y)$ , so  $P_{\pm}(\mu)^2$  are roots of  $Q(\mu, \cdot)$  for  $\mu < 1$ . For  $\mu = 1$ ,  $P_{\pm} = \sqrt{2}$  by Remark 2 below. For  $0 < \mu < 1$  both are positive roots, hence each lies in  $\{y_1(\mu), y_2(\mu)\}$ . The three roots of  $G$  are simple, so  $P_+(\mu)$  and  $P_-(\mu)$  depend continuously on  $\mu$ , and so do  $y_1(\mu) < y_2(\mu)$ . By connectedness each of  $P_+^2$  and  $P_-^2$  coincides with the same  $y_i$  on all of  $(0, 1)$ . At  $\mu = 1/2$  the proof of Proposition 1, which uses only (a) and the first part of (b), gives certified enclosures  $P_-^2 \in [0.99, 1.30]$  and  $P_+^2 \in [4.70, 5.02]$ , each containing exactly one root of  $Q(1/2, \cdot)$ ; so  $P_-^2 = y_1$  and  $P_+^2 = y_2$ .

(c) For  $0 < \mu < 1$  the roots  $y_i(\mu)$  are simple, hence differentiable, and  $y'_i = -(\partial Q / \partial \mu) / (\partial Q / \partial y)$ . A zero of  $y'_1$  at some  $\mu$  would be a common root of  $Q(\mu, \cdot)$  and  $(\partial Q / \partial \mu)(\mu, \cdot)$ , but their resultant with respect to  $y$ ,

$$7044820107264 \mu^7 (\mu - 1)^3 (\mu + 1)^7 (\mu + 2)^3 (2\mu + 1)^3 R^6 E B^3,$$

where  $E = 4\mu^6 + 12\mu^5 + 21\mu^4 + 22\mu^3 + 21\mu^2 + 12\mu + 4$ , has no zero in  $(0, 1)$ . So  $y_1$  is strictly monotone on  $(0, 1)$ . The leading coefficient of  $Q(\mu, \cdot)$  does not vanish at  $\mu = 1$ , so the roots stay bounded as  $\mu \rightarrow 1^-$ , and  $y_1$  tends to a nonnegative root of  $Q(1, \cdot)$ , which is 2. Since  $y_1(1/2) = 1.1335 \dots < 2$ ,  $y_1$  is increasing, and therefore  $y_1(\mu) < 2 = P_-(1)^2$  for  $\mu < 1$ . As  $\mu \rightarrow 0^+$  it therefore decreases to a limit  $L \geq 0$ , which is a root of  $Q(0, y) = -16(4y - 3)$ , so  $L = 3/4$ .  $\square$

**Corollary 1.** *Every self-similar collapse of three point vortices has  $|\omega_0|t_c > \sqrt{3}/2$ , and for every  $\delta > 0$  there is one with  $|\omega_0|t_c < \sqrt{3}/2 + \delta$ . Equivalently, each vortex travels more than twice its initial distance from the collision point, and the factor 2 is sharp.*

*Proof.* By Lemma 2, the normalization (4) and Theorem 1,  $P \geq P_-(\mu) > \sqrt{3}/2$ , and  $P_-(\mu) \rightarrow \sqrt{3}/2$  as  $\mu \rightarrow 0^+$ . The path length is  $r_0 \sqrt{1 + 4P^2} > 2r_0$ , and no vortex starts at the collision point, since  $z_1, z_2$  and  $z_3$  in (5) never vanish.  $\square$

Every collapse of three point vortices in the plane is self-similar [8, Theorem 1.1], [9, Prop. 5.2], [18, Sect. 3.1], and every motion of three vortices that ends in a total collision is self-similar [13, Theorem 1], [14, Sect. 3], so the bound of Corollary 1 holds for every collapse of three point vortices. The bound does not carry over to larger systems: the collapsing configuration of seven vortices in [7, Table 1, Fig. 1a] has the constant  $\Omega = 12433/9000 - (31\sqrt{155}/450)i$ , where  $\Omega = 2\pi i \bar{\kappa}$  [7, Eq. (11)], so  $P = |\operatorname{Re} \Omega| / (2|\operatorname{Im} \Omega|) = 12433 / (1240\sqrt{155}) = 0.8053 \dots < \sqrt{3}/2$ .

The two minima are shown in Figure 1.

### 3.3 The ratio $\mu = 1/2$

For  $\Gamma = (1, 1/2, -1/3)$  we have  $R = 7/4$  and  $\cos \theta_0 = -\sqrt{7}/14$ , and (7) becomes

$$P(\theta) = \frac{14 \sin^2 \theta + 6\sqrt{7} \cos \theta + 21}{2(14 \cos \theta + \sqrt{7}) \sin \theta}.$$

**Proposition 1.** *Let  $X = 245351/5201^{3/2}$  and  $\sigma = 7\sqrt{5201}/162$ . For  $\mu = 1/2$ ,*

$$P_- = \sqrt{\frac{605}{324} + \sigma \cos\left(\frac{1}{3} \arccos X - \frac{2\pi}{3}\right)} = 1.0647059762712043 \dots,$$

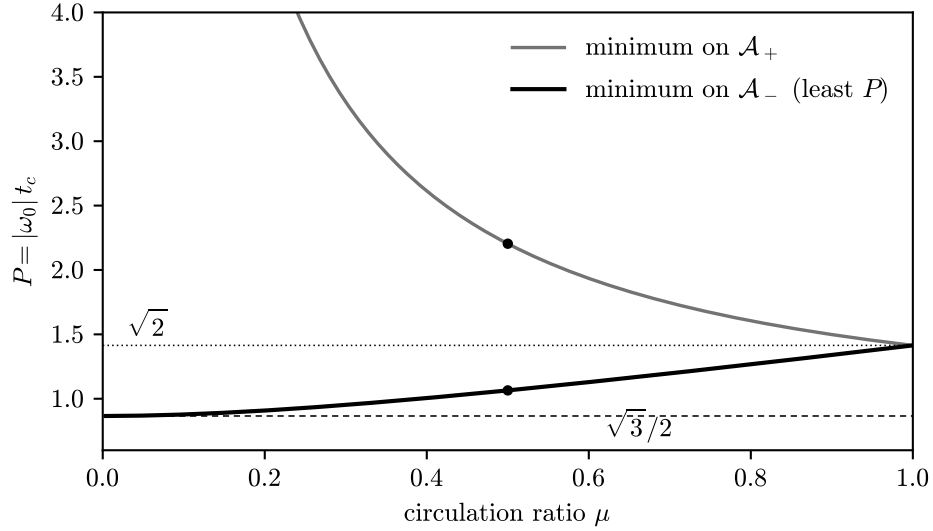


Figure 1: The minima  $P_-(\mu)$  and  $P_+(\mu)$  of  $P = |\omega_0|t_c$  on the two collapsing arcs, as functions of the circulation ratio  $\mu$ . The dots mark  $\mu = 1/2$  (Proposition 1). As  $\mu \rightarrow 0$ ,  $P_-$  tends to  $\sqrt{3}/2$  and  $P_+$  grows without bound.

$$P_+ = \sqrt{\frac{605}{324} + \sigma \cos\left(\frac{1}{3} \arccos X\right)} = 2.2038550160361327 \dots,$$

attained at  $\cos \theta = -0.9243893679 \dots$  and  $\cos \theta = 0.6739838839 \dots$  respectively. Both are roots of the polynomial  $8748\xi^6 - 49005\xi^4 + 27794\xi^2 + 18723$ , which is irreducible over  $\mathbb{Q}$  and whose real roots are exactly these two numbers and their negatives.

*Proof.* Here  $16Q(1/2, \xi^2) = 8748\xi^6 - 49005\xi^4 + 27794\xi^2 + 18723$ , and by Theorem 1(b) the squares of both minima are roots of this polynomial. As a cubic in  $q = \xi^2$  it is irreducible over  $\mathbb{Q}$  and has positive discriminant, so its roots are real and are given by the trigonometric solution of the cubic,  $q_m = 605/324 + \sigma \cos(\frac{1}{3} \arccos X - 2\pi m/3)$ ,  $m = 0, 1, 2$ , with  $q_0 \approx 4.856977$ ,  $q_1 \approx 1.133599$  and  $q_2 \approx -0.388724$ . The critical points are the roots of  $G$  in the two intervals of the proof of Theorem 1(a); with  $C = (\sqrt{7}/2) \cos \theta$  they are at the stated values of  $\cos \theta$ . Exact sign changes of  $G$  place them in  $0.67 < \cos \theta < 0.68$  on  $\mathcal{A}_+$  and  $-0.93 < \cos \theta < -0.92$  on  $\mathcal{A}_-$ , and interval arithmetic applied to  $P^2 = N^2/(4\mu^2(R - C^2)M^2)$  on these intervals gives  $P_+^2 \in [4.70, 5.02]$  and  $P_-^2 \in [0.99, 1.30]$ . Each enclosure contains exactly one of  $q_0, q_1, q_2$ , so  $P_+^2 = q_0$  and  $P_-^2 = q_1$ .  $\square$

**Remark 1.** The two minima are conjugate algebraic numbers of degree six. Since the cubic in  $q$  is irreducible over  $\mathbb{Q}$  and has three real roots, none of its roots is expressible by real radicals (casus irreducibilis), and therefore neither minimum is. The same holds for every rational  $\mu = a/b \in (0, 1)$  with  $b \leq 30$ : for each of these 277 values,  $Q(\mu, \xi^2)$  is irreducible over  $\mathbb{Q}$ .

**Remark 2.** For  $\mu = 1$ , that is  $\Gamma = (1, 1, -1/2)$ , the zero-impulse circle is  $|w - 1/2| = \sqrt{3}/2$ . With  $w = 1/2 + (\sqrt{3}/2)e^{i\beta}$ ,  $z_1 = 0$  and  $z_2 = 1$ , the same computation gives  $P = (3 - \cos 2\beta)/(2 \sin 2\beta)$  on the collapsing arcs  $0 < \beta < \pi/2$  and  $\pi < \beta < 3\pi/2$ , which are exchanged by interchanging the two equal vortices ( $w \mapsto 1 - w$ ). Since  $P$  has period  $\pi$  in  $\beta$ , Lemma 1 applies on both arcs and gives  $P \geq \sqrt{2}$ , with equality if and only if  $\cos 2\beta = 1/3$ , in agreement with Theorem 1. This parametrization is Kimura's [15, Sect. 4], who gives the two rates for  $\Gamma = (2, 2, -1)$  in his Eq. (4.4);

their ratio  $B/(-2A)$  is this expression for  $P$ , which is also a specialization and reparametrization of Gröbli's spiral coefficient [11, Sect. 10].

**Remark 3.** Interchanging the first two vortices and multiplying the circulations by  $1/\mu$  maps the family with ratio  $\mu$  onto the family with ratio  $1/\mu$  and exchanges the two orientations, so the arc minima satisfy  $P_+(\mu) = P_-(1/\mu)$ . For  $\mu > 1$  the smaller minimum is therefore attained on  $\mathcal{A}_+$ .

## 4 Two concentric regular polygons

In this section  $\theta$  denotes the relative rotation of two rings. Let  $n \geq 2$  and  $\epsilon = e^{2\pi i/n}$ , and place vortices of circulation  $x > 0$  at  $z\epsilon^k$  and vortices of circulation  $-1$  at  $\zeta\epsilon^k$ ,  $k = 0, \dots, n-1$ . The motion preserves this symmetry, and the center of vorticity is the origin. Using

$$\sum_{k=1}^{n-1} \frac{1}{1-\epsilon^k} = \frac{n-1}{2}, \quad \sum_{k=0}^{n-1} \frac{1}{z-\zeta\epsilon^k} = \frac{nz^{n-1}}{z^n-\zeta^n},$$

one finds that (1) reduces to [2, Eq. (3)]

$$\bar{z} = \frac{1}{2\pi i} \left( \frac{x(n-1)}{2z} - \frac{nz^{n-1}}{z^n-\zeta^n} \right), \quad \bar{\zeta} = \frac{1}{2\pi i} \left( -\frac{n-1}{2\zeta} + \frac{xn\zeta^{n-1}}{\zeta^n-z^n} \right). \quad (10)$$

A collapse requires the angular impulse  $n(x|z|^2 - |\zeta|^2)$  to vanish, and after a rotation and a dilation we take  $z = 1$  and  $\zeta = \sqrt{x} e^{i\theta}$ . The motion is self-similar exactly when  $\dot{z}/z = \dot{\zeta}/\zeta$ . Writing  $v = \zeta^n$ , (10) gives

$$\frac{\dot{z}}{z} = \frac{1}{2\pi i} \left( \frac{x(n-1)}{2} - \frac{n}{1-v} \right), \quad \frac{\dot{\zeta}}{\zeta} = \frac{1}{2\pi i} \left( -\frac{n-1}{2x} - \frac{nv}{1-v} \right),$$

and these are equal if and only if

$$(n-1)x^2 - 2nx + (n-1) = 0. \quad (11)$$

This is the circulation condition of Koiller et al. [16, Sect. 11]. Demina and Kudryashov [7, Sect. 3] place circulations  $\Gamma_1$  and  $\Gamma_2 = -\Gamma_1/r^2$  on two such rings, where  $r$  is the ratio of the radii, and a vortex of circulation  $\Gamma_0$  at the center; for  $\Gamma_0 = 0$  their equation for  $r$  [7, Eq. (37)] is (11) with  $x = r^2$ . The condition (11) coincides with the classical condition  $\sum_{i < j} \Gamma_i \Gamma_j = 0$  for the  $2n$  vortices, whose left side here equals  $(n/2)((n-1)x^2 - 2nx + (n-1))$ . Its roots are  $x_n$  and  $1/x_n$ , where

$$x_n = \frac{n + \sqrt{2n-1}}{n-1} = e^\eta, \quad \cosh \eta = \frac{n}{n-1}.$$

The root  $1/x_n$  gives the same family with the two rings interchanged, as one sees by multiplying the circulations by  $-x_n$  and reflecting the configuration, so we take  $x = x_n$ . Then every  $\theta$  gives a self-similar motion. Let  $\rho = x_n^{n/2} = e^{n\eta/2} > 1$  and  $\alpha = n\theta$ , so that  $v = \rho e^{i\alpha}$ . The common value of  $\bar{\kappa}$  is  $S/(2\pi i)$  with  $S = x_n(n-1)/2 - n/(1-v)$ , so  $\kappa = (\text{Im } S + i \text{Re } S)/(2\pi)$ , and since  $\text{Im}(1-v)^{-1} = \rho \sin \alpha / |1-v|^2$ ,

$$\text{Re } \kappa = -\frac{n\rho \sin \alpha}{2\pi |1-v|^2}.$$

The configuration therefore collapses exactly when  $\sin n\theta > 0$ , that is, for  $0 < \theta < \pi/n$  modulo  $2\pi/n$ . Using  $|1-v|^2 = 1 - 2\rho \cos \alpha + \rho^2$  one finds

$$\frac{|1-v|^2 \text{Re } S}{\rho} = K_n - ((n-1)x_n - n) \cos \alpha, \quad K_n = \frac{(n-1)x_n(\rho + \rho^{-1})}{2} - \frac{n}{\rho}.$$

Here  $(n-1)x_n - n = \sqrt{2n-1}$ , and writing  $n = (n-1) \cosh \eta$  in the last term of  $K_n$  gives

$$K_n = (n-1) \sinh \frac{(n+2)\eta}{2}.$$

Moreover

$$K_n - \sqrt{2n-1} = \frac{(n-1)x_n(\rho^{1/2} - \rho^{-1/2})^2}{2} + n(1 - \rho^{-1}) > 0,$$

so  $\text{Re } S > 0$ , and (2) gives

$$P = \frac{K_n - \sqrt{2n-1} \cos n\theta}{2n \sin n\theta}, \quad 0 < n\theta < \pi. \quad (12)$$

The collapse rate and the rotation rate as functions of the relative angle are given by Koiller et al. [16, Sect. 11]. The constant  $S$  is the constant  $\Omega$  of [7, Eq. (11)], and Demina and Kudryashov write it for these rings, with the central vortex, as an explicit function of  $e^{in\theta}$  [7, Eq. (36)]; for  $\Gamma_0 = 0$ ,  $\Gamma_1 = x_n$  and radii 1 and  $\sqrt{x_n}$  their expression reduces to  $S$  by (11). (12) writes the ratio of the two rates in closed form.

**Proposition 2.** *For  $n \geq 2$  the collapsing configurations of two concentric regular  $n$ -gons with circulations  $x_n$  and  $-1$  satisfy*

$$P \geq F_n = \frac{\sqrt{K_n^2 - (2n-1)}}{2n},$$

*with equality if and only if  $\cos n\theta = \sqrt{2n-1}/K_n$ .*

*Proof.* Apply Lemma 1 to (12) with  $a = K_n$  and  $b = \sqrt{2n-1}$ . □

For small  $n$  the constants are as follows.

| $n$ | $x_n$              | $K_n$           | $F_n$             | $\cos n\theta$ at the minimum |
|-----|--------------------|-----------------|-------------------|-------------------------------|
| 2   | $2 + \sqrt{3}$     | $4\sqrt{3}$     | $3\sqrt{5}/4$     | $1/4$                         |
| 3   | $(3 + \sqrt{5})/2$ | 11              | $\sqrt{29}/3$     | $\sqrt{5}/11$                 |
| 4   | $(4 + \sqrt{7})/3$ | $55\sqrt{7}/9$  | $\sqrt{322}/9$    | $9/55$                        |
| 5   | 2                  | $127\sqrt{2}/8$ | $\sqrt{31682}/80$ | $12\sqrt{2}/127$              |

For  $n = 2$  the configuration is a parallelogram of four vortices; four-vortex collapse is treated by Novikov and Sedov [21, Sect. 4]. The ratio  $x_n$  is rational exactly when  $2n-1$  is a perfect square, and  $n = 5$  is the smallest such  $n$ . There  $x_5 = 2$  and

$$P = \frac{127\sqrt{2} - 24 \cos 5\theta}{80 \sin 5\theta} \geq \frac{\sqrt{31682}}{80} = 2.2249297741726591 \dots$$

## 5 Numerical verification

The results were checked numerically as summarized in Table 1. Except where the table says otherwise, the checks evaluate the Biot–Savart velocities (1) directly for configurations built independently of (5) (two vortices fixed and the third moved around the zero-impulse circle), test self-similarity by comparing  $\dot{z}_j/(z_j - z_c)$  across all vortices, and obtain  $P$  from (2). The algebraic steps in the proofs were verified in exact arithmetic (SymPy): Lemma 3 for all three vortices, (9), the resultant in the proof of Theorem 1(b), the discriminant and the resultant in the proof of Theorem 1(c), the irreducibility of  $Q$  and of each of the 277 sextics in Remark 1, the constants in Proposition 1, and, for general  $n$ , the identities of Section 4 from (11) to (12), together with the table of constants.

| Check                                                                                                                                                                                                                 | Result                                                                                                                                                                                                                               |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Lemma 3 against the Biot–Savart velocities of all three vortices, at the positions (5) for 36 pairs $(\mu, \theta)$ on a grid with $0.1 \leq \mu \leq 7$ , and for 30 triangles built independently of (5), 50 digits | relative difference $\leq 2 \times 10^{-49}$                                                                                                                                                                                         |
| Arc minima by direct minimization of $P$ , computed from the quotient of the first vortex, for 26 values of $\mu \in (0, 1]$ at 30 digits and 13 rational values at 50 digits                                         | agree with the roots of $Q(\mu, \cdot)$ to $5 \times 10^{-21}$ and $5 \times 10^{-45}$                                                                                                                                               |
| $\mu = 1/2$ : the three quotients over 4000 shapes on the zero-impulse circle, 60 digits                                                                                                                              | equal to 60-digit working precision; the collapsing shapes fill half of the circle, the arcs $\mathcal{A}_\pm$                                                                                                                       |
| $\mu = 1/2$ : local minima of $P$ over the collapsing shapes, located numerically                                                                                                                                     | two, equal to the values of Proposition 1 to 60-digit working precision                                                                                                                                                              |
| $\mu = 1/2$ : formula for $P(\theta)$ against direct evaluation at the positions (5), 399 interior points of each arc                                                                                                 | relative difference $\leq 10^{-59}$                                                                                                                                                                                                  |
| $\mu = 1/2$ : integration of (1) from the minimizing configuration on $\mathcal{A}_-$ (Taylor method, 30 digits)                                                                                                      | every $ z_j - z_c ^2$ follows $1 - t/t_c$ , and every rotation angle follows $-\omega_0 t_c \ln(1 - t/t_c)$ , to 30-digit working precision up to $t = 0.9 t_c$                                                                      |
| Five configurations at the positions (5) with $\mu \in \{0.3, 0.5, 2\}$ , on both arcs: integration of (1) to $t = 0.9 t_c$ (fourth-order Runge–Kutta, 30 digits)                                                     | distances to the collision point, rotation angles, the angle $\arctan 2P$ between each velocity and the direction to the collision point, and path lengths $r_0 \sqrt{1 + 4P^2} (1 - \sqrt{1 - t/t_c})$ agree to $2 \times 10^{-10}$ |
| $\mu = 1$ (Remark 2): $P$ from (1) against $(3 - \cos 2\beta)/(2 \sin 2\beta)$ on both collapsing arcs, and its minimum, 50 digits                                                                                    | difference $\leq 2 \times 10^{-47}$ ; minimum $\sqrt{2}$ at $\cos 2\beta = 1/3$ on both arcs                                                                                                                                         |
| $\mu = 1$ : the configuration with the fastest collapse at $ z_1 - z_2  = 1$ , 50 digits                                                                                                                              | $\cos 2\beta = 3/5$ , $t_c = 4\pi/3$ and $P = 3/2$ to $10^{-40}$                                                                                                                                                                     |
| $\mu = 1$ : Kimura’s rates $A, B$ [15, Eq. (4.4)] at ten angles in $0 < \beta < \pi/2$ , 50 digits                                                                                                                    | $\kappa = (A + iB)/(4\pi)$ for $\Gamma = (1, 1, -1/2)$ , to $10^{-45}$                                                                                                                                                               |
| Two rings, $n = 2, \dots, 10$ : the reduced equations (10) and the two sums over roots of unity against the full Biot–Savart sum, 50 digits                                                                           | difference $\leq 10^{-49}$                                                                                                                                                                                                           |
| Two rings, $n = 2, \dots, 8$ : $P$ from (1) against (12) over the collapsing range, and its numerical minimum against $F_n$ , 60 digits                                                                               | relative difference $\leq 3 \times 10^{-58}$ ; minima agree to $2 \times 10^{-60}$                                                                                                                                                   |
| Two rings: the equations of Demina and Kudryashov [7, Eqs. (36)–(37)] at $\Gamma_0 = 0$ (exact, SymPy), and their constant against the Biot–Savart sum, $n = 2, \dots, 8$ , 50 digits                                 | (37) is (11) with $x = r^2$ ; (36) minus $S$ is a multiple of (11); difference $\leq 10^{-45}$                                                                                                                                       |
| The collapsing configuration of seven vortices in [7, Table 1, Fig. 1a], Biot–Savart at 50 digits                                                                                                                     | self-similar, the printed $\Omega$ to $10^{-45}$ , and $P = 12433/(1240\sqrt{155}) < \sqrt{3}/2$                                                                                                                                     |

Table 1: Numerical checks of the results.

## 6 Discussion

Lemma 3 gives, in a form suited to minimization, the collapse and rotation rates that Aref [3, Eqs. (25a), (25d)] expresses through the side lengths of the triangle (see also [10], [9, Eq. (E.1)], and, for equal circulations, [15, Eq. (4.4)]); Aref also notes that the product of the two rates fixes the logarithmic spiral [3, Eq. (29c)]. Theorem 1 concerns the minimization of that product, which neither [3], [6] nor [15] considers. When the two circulations of the same sign differ,  $\mu < 1$ , the two orientations of the triangle are not equivalent and the minima on the two arcs differ, so a minimization restricted to one orientation finds only one of them. The infimum  $\sqrt{3}/2$  of Corollary 1 is approached only as  $\mu \rightarrow 0$  and is not attained.

Leoncini, Kuznetsov and Zaslavsky [19] analyze the motion near collapse when two of the vortices are identical; the configuration that collapses fastest at a fixed distance between the identical vortices, found by Kimura [15, Eq. (4.6)] and shown in [19, Fig. 18], has  $P = 3/2$ , and the value  $\sqrt{3}/2$  quoted there is that of their energy parameter  $\Lambda = e^{4\pi H}$ , not of  $P$ . For  $\sum_{j < k} \Gamma_j \Gamma_k = 0$ , Tavantzis and Ting [25, Sect. II], who work with the side lengths of the triangle, show that the contracting family of similar configurations is unstable and the expanding family asymptotically stable, stability referring to the ratios of the sides. Reinaud, Dritschel and Scott [23] extend the collapse conditions to generalized Euler and quasi-geostrophic models and map the collapse time over the collapsing configurations, with the distance between the like-signed vortices fixed, finding local minima of that time;  $P$ , unlike the collapse time, does not depend on the choice of length and time scales. Krishnamurthy and Stremler [18, Sects. 3.3–3.5] relate the interior angles of the triangle, the circulation ratios, the energy, the collapse time and the distance traveled before collapse by the circumcenter of the triangle. When the angular impulse about the center of vorticity vanishes, the circumcircle passes through the center of vorticity at all times [17, Eq. (40)], so in a collapse the circumcenter stays one circumradius from the collision point; since it is carried by the self-similar motion, it travels  $\sqrt{1 + 4P^2}$  times the initial circumradius before the collapse. In their notation  $P = \tilde{\tau}|K_1|$  [18, Eq. (3.28a)], and they observe numerically that this normalized distance, their  $\tilde{s}(1)$ , exceeds 2 [18, Sect. 3.5]; Corollary 1 proves this observation and shows that the constant 2 cannot be improved.

Gallay and Šverák [9] ask when a collision of three point vortices is regularizable, that is, a limit of nearby motions without collision. Whether it is depends on the perturbation [9, Theorems 5.8, 5.10, 5.15], and when it is, the limit is determined only up to a rotation [9, Sect. 1], because the triangle turns through the angle  $s \ln(1 - t/T)$  and so makes infinitely many turns before the collision [9, Sect. 5.2]. Grotto, Romito and Viviani [12] select a continuation after collapse by adding a vanishing stochastic diffusion, which yields a probability distribution over continuations rather than a single one. They note that  $s \neq 0$  [9, Remark E.2]; since  $|s| = P$ , Corollary 1 sharpens this to  $|s| > \sqrt{3}/2$  for every collision, and no larger constant holds for all of them.

For two rings, Aref [2] derived the reduced equations (10) for arbitrary circulations and analyzed equal and opposite circulations,  $x = 1$ , which do not satisfy (11). Koiller et al. [16] found the collapsing configurations and their rates. Demina and Kudryashov [7] study relative equilibria, collapse and scattering of point vortices with arbitrary circulations by a polynomial method. For two regular polygons with a vortex of circulation  $\Gamma_0$ , possibly zero, at the center, they give the equation for the ratio of the radii, which for  $\Gamma_0 = 0$  is (11), and the constant  $\Omega$ , which for  $\Gamma_0 = 0$  is  $S$ , as an explicit function of the relative rotation [7, Eqs. (36)–(37)], and they state that every relative rotation with  $e^{in\theta} \neq \pm 1$  gives a collapse or a scattering. For this family they do not say which of the two occurs, separate the collapse rate from the rotation rate, or evaluate their ratio. (12) writes this ratio in closed form, and Proposition 2 gives its minimum over the relative rotation. O’Neil [22] proves that for generic circulations three concentric rings have finitely many relative equilibria and

collapse configurations. We have not found the minimal values of Theorem 1 and Propositions 1 and 2 stated in the literature, nor a proof of the bound of Corollary 1 or of its sharpness.

**Data availability.** The programs `verify_general_mu.py`, `verify_floors_independent.py` and `plot_minimal_winding.py`, used for Section 5 and Figure 1, and their output are in the repository <https://github.com/ChaseHendrick/minimal-winding>, the programs in `code/` and their output in `data/`.

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